Panoptic MapAnything

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PLR Lab WS-2025

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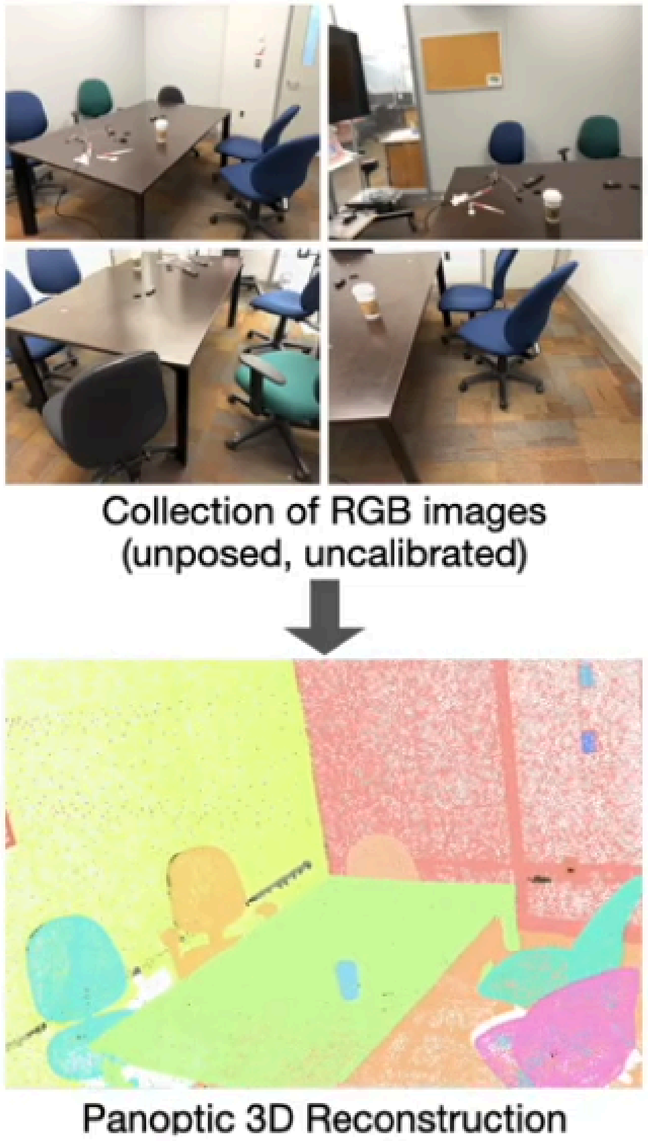
[1] Bowen Cheng et al. 'Masked-attention Mask Transformer for Universal Image Segmentation', CVPR 2022
[2] Nikhil Keetha et al. 'MapAnything: Universal Feed-Forward Metric 3D Reconstruction', CVPR 2025
[3] René Ranftl et al. 'Vision Transformers for Dense Prediction', ICCV 2021

Project Overview [1] [2] [3]

- **Problem**: Current 3D panoptic scene segmentation approaches:
  - Require specific inputs (point clouds, posed RGB images)
  - Demand costly test-time optimization for 2D-3D alignment
  - Include post-processing steps which add additional overhead
  - Separate segmentation and reconstruction pipelines ➜ efficiency losses

- **Opportunity**: Foundation models like MapAnything enable fast, metric-accurate 3D modeling

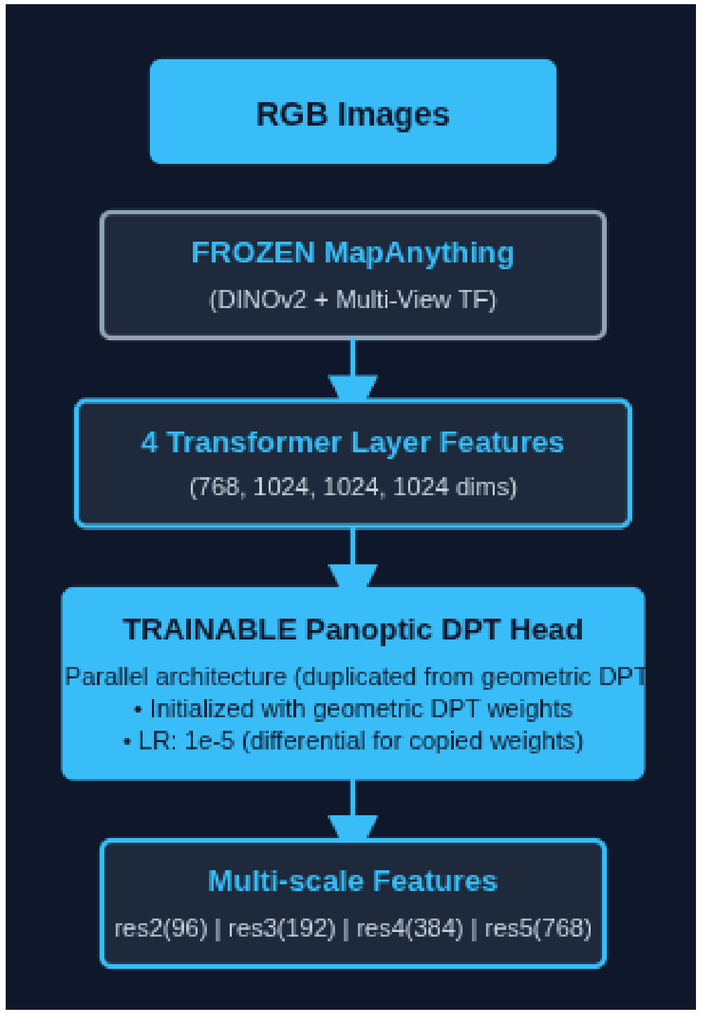
- **Goal**: Enhance MapAnything with multi-view panoptic segmentation to create unified, feature-rich semantic 3D maps



[1] Bowen Cheng et al. 'Masked-attention Mask Transformer for Universal Image Segmentation', CVPR 2022
[2] Nikhil Keetha et al. 'MapAnything: Universal Feed-Forward Metric 3D Reconstruction', CVPR 2025
[3] Lojze Žust et al. 'PanSt3R: Multi-view Consistent Panoptic Segmentation', ICCV 2025

Our Approach[1] [2] [3]

Step 1: Extracting 2D & 3D Features



Harnesses **MapAnything**'s pre-trained feature extractors:

- **DINOv2** encoder → 2D features (768-dim)
- **Multi-View Transformer** → 3D features (1024-dim)

Trainable Parallel DPT Head:

- Reassemble & Projections
- Resize Layers (4× up/2× up/identity/2× down)
- RefinNet Fusion (bottom-up)
- Output Projections (1×1 conv)

Multi-scale Output:

- res2: H/4 (96ch) | res3: H/8 (192ch)
- res4: H/16 (384ch) | res5: H/32 (768ch)

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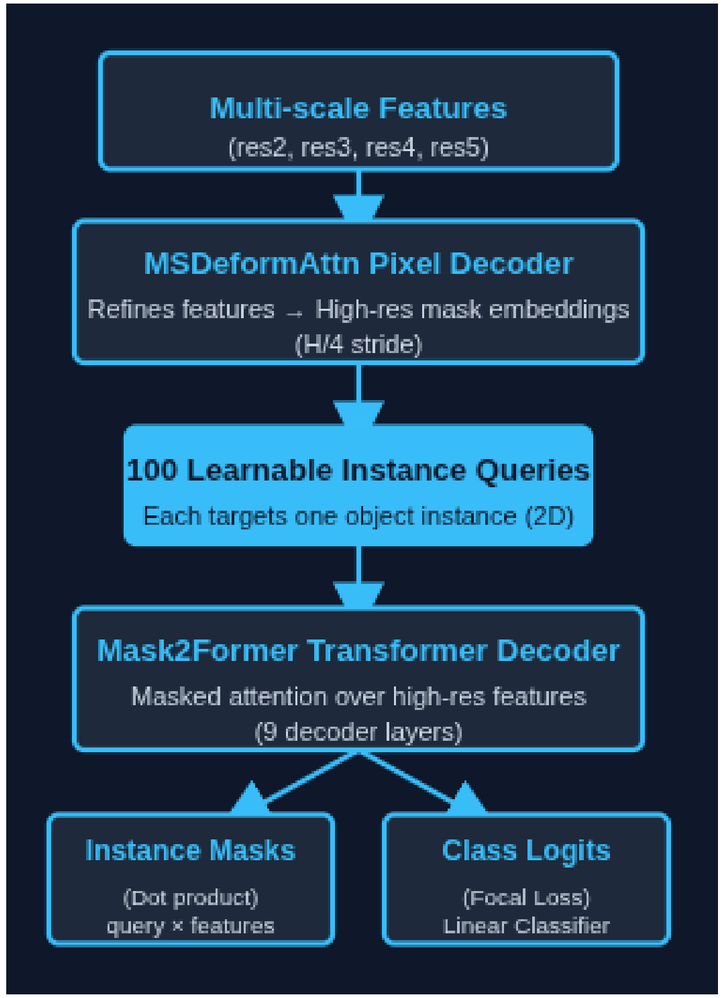
[2] Nikhil Keetha et al. 'MapAnything: Universal Feed-Forward Metric 3D Reconstruction', CVPR 2025
[3] Image generated using Perplexity (perplexity.ai)

Our Approach[1] [2] [3]

Step 2: Panoptic Predictions



Pixel Decoder

- MSDeformAttn (6 layers) refines multi-scale features

Transformer Decoder

- 9 masked-attention layers with 100 instance queries

Prediction Heads

- Instance masks (dot product) + class logits (linear classifier + focal loss)

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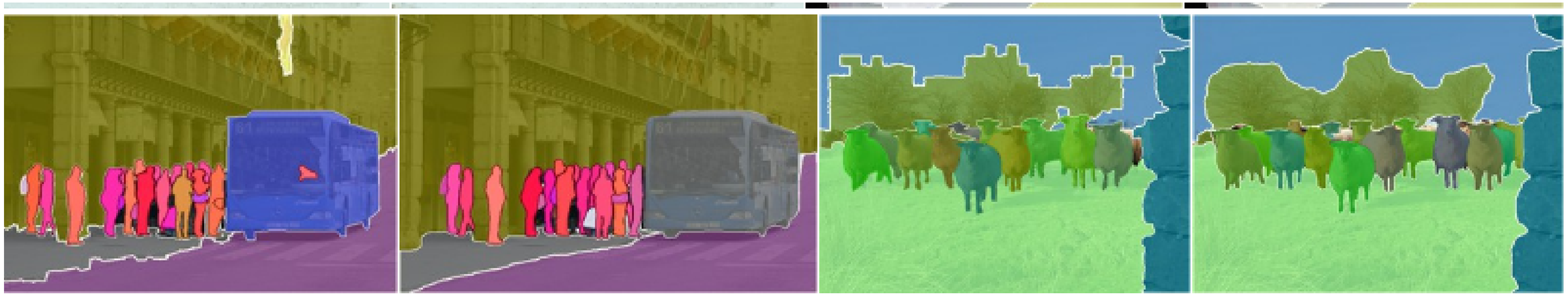
[2] Nikhil Keetha et al. 'MapAnything: Universal Feed-Forward Metric 3D Reconstruction', CVPR 2025
[3] Image generated using Perplexity (perplexity.ai)

Training Details (1/3)

Dataset

- **COCO Panoptic 2017** (2D, real, 118K train images, 5K val images, 133 classes)
  a. Things: 80 classes (person, car, dog, etc.)
  b. Stuff: 53 classes (sky, grass, wall, etc.)



Visualization of panoptic segmentation predictions on the COCO panoptic dataset: Mask2Former with Swin-L backbone. First and third columns: ground truth. Second and fourth columns: prediction.

Training Details (2/3)

Loss Function

1. Matching

- Find optimal matching between GT instances and predicted instances (queries) - Hungarian matching
- Unmatched queries supervised as "no object"

2. Loss Components

- Instance mask loss: binary cross entropy + dice loss
- Instance classification loss: focal loss (uses all 133 classes in COCO panoptic)

$$L = \lambda_{cls} L_{cls} + \lambda_{mask} L_{mask} + \lambda_{dice} L_{dice}$$

3. Loss Function Configuration

Loss Weights:

- Classification loss (focal): 5.0
- Mask loss (BCE + Dice): 5.0 each
- No-object weight: 0.1 (Hungarian matching)

```
[12/14 18:41:29 d2.utils.events]:  eta: 0:00:12  iter: 89979  total_loss: 44.11  loss_ce: 1.697  loss_mask: 0.5985  loss_dice: 1.492  loss_ce_0: 8.97  loss_
mask_0: 0.6826  loss_dice_0: 1.809  loss_ce_1: 2.352  loss_mask_1: 0.658  loss_dice_1: 1.688  loss_ce_2: 2.163  loss_mask_2: 0.624  loss_dice_2: 1.65  loss_
ce_3: 1.988  loss_mask_3: 0.6078  loss_dice_3: 1.609  loss_ce_4: 1.802  loss_mask_4: 0.6059  loss_dice_4: 1.574  loss_ce_5: 1.791  loss_mask_5: 0.5975  loss
_dice_5: 1.571  loss_ce_6: 1.747  loss_mask_6: 0.5991  loss_dice_6: 1.56  loss_ce_7: 1.731  loss_mask_7: 0.5974  loss_dice_7: 1.494  time: 0.6110  data_time
: 0.0256  lr: 1e-05  max_mem: 20468M
[12/14 18:41:41 d2.engine.hooks]: Overall training speed: 89998 iterations in 15:17:59 (0.6120 s / it)
[12/14 18:41:41 d2.engine.hooks]: Total training time: 18:02:36 (2:44:36 on hooks)
```

Snapshot from the slurm training logs showing the total loss is composed of multiple loss components

Training Details (3/3)

Training Configuration

Learning Setup

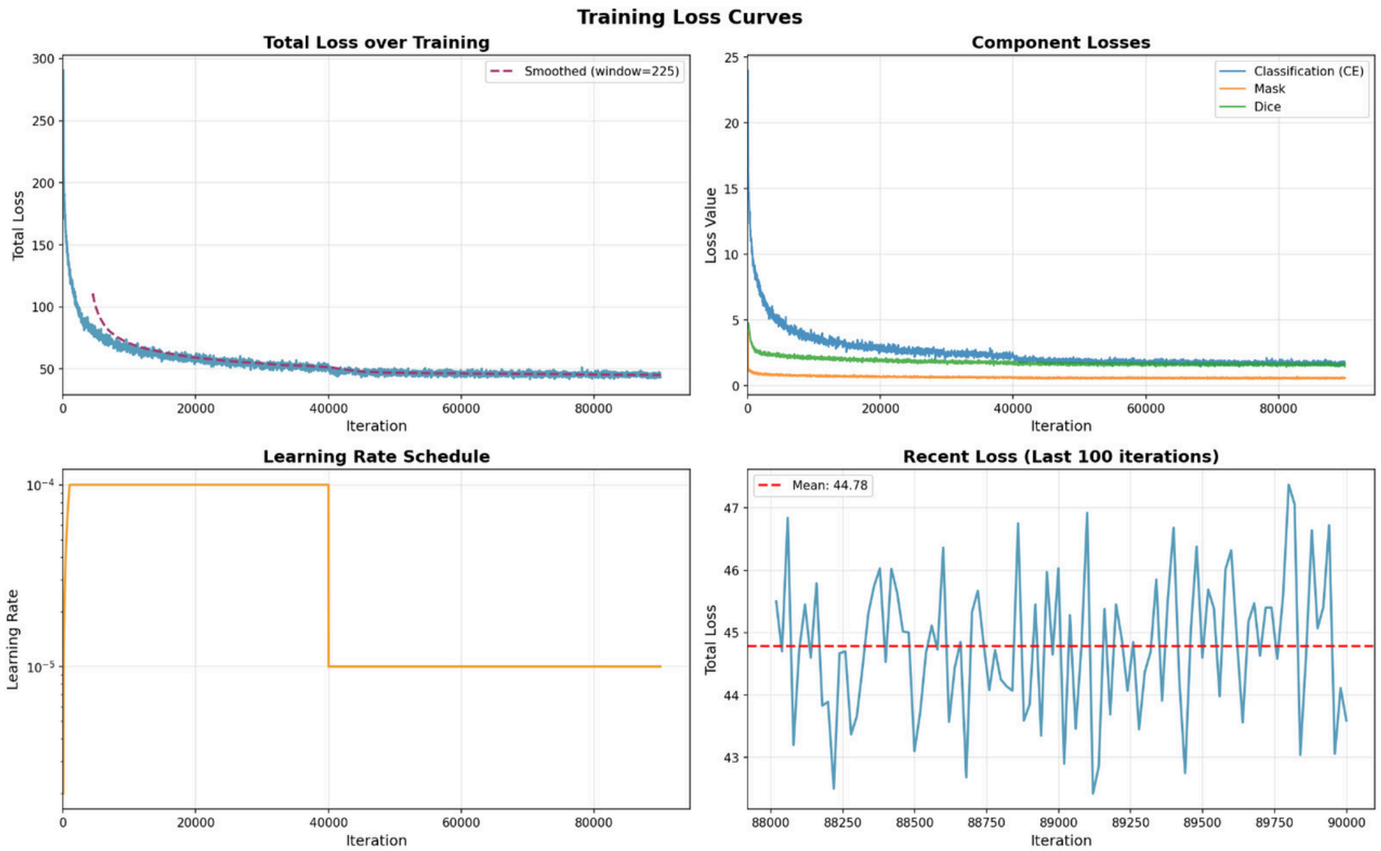
- Optimizer: AdamW (weight decay: 0.05, L2 clip: ≤1.0)
- Differential Learning Rates:
  - Panoptic DPT: 1e-5 (29M params)
  - DPT Projections: 1e-4
  - Mask2Former Head: 1e-4

Learning Rate Schedule

- Max Iterations: 50,000
- Warmup: 1,000 iterations (linear, 0.001×)
- Decay: 0.1× at iteration 40,000

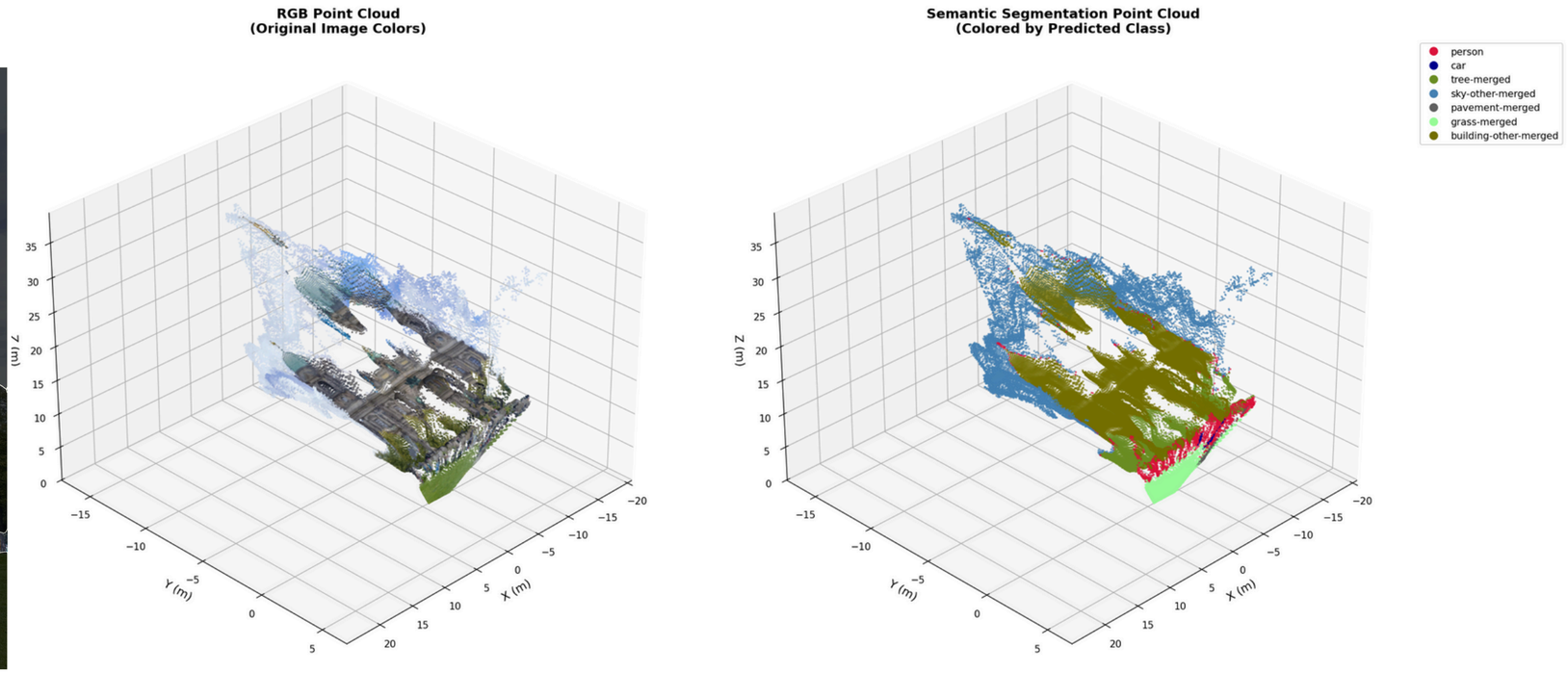
Post-Processing

- Confidence threshold: 0.8 (filters uncertain predictions)
- Overlap threshold: 0.8 (removes duplicate detection via NMS)



Training Results

Panoptic Segmentation result from the trained model for the input image.



Input image lifted to 3D point cloud using MapAnything backbone, with output from Panoptic Head projected onto the 3D point cloud

Evaluation against benchmark[1]

Our Model is currently trained for single view loss → Benchmark for our model is pre-trained **Mask2former with ResNet/Swin-L backbones.**

We validate the trained model against COCO val2017 (5K images) using the COCOPanopticEvaluator.

Three metrics track performance :-
Panoptic Quality (PQ) → measures combined detection and segmentation accuracy
Segmentation Quality (SQ) → assesses mask boundary precision
Recognition Quality (RQ) → evaluates instance detection correctness.

Method	Backbone	PQ	PQ Things	PQ Stuff
Mask2Former (our baseline)	MapAnything + DPT adapter	40	42.2	36.8
Mask2Former [1]	R50	51.9	57.7	43
Mask2Former [1]	R101	52.6	58.5	43.7
Mask2Former [1]	Swin-L	57.8	64.2	48.1

[1] Bowen Cheng et al. 'Masked-attention Mask Transformer for Universal Image Segmentation', CVPR 2022

Conclusion & Next Steps

- Setup our training environment on Marvin
- Modify our architecture such that we can train on 3D datasets
- Sample **ScanNet++**[1] with nearby multi-view poses for overlapping field of view
- Develop **multi-view consistency loss** for panoptic segmentation alignment
- Experiment with **SigLip/SigLip2** for open-vocabulary capabilities (if time permits)

[1] Chandan Yeshwanth et al. ‘ScanNet++: A High-Fidelity Dataset of 3D Indoor Scenes’, ICCV 2023